

Resistance Atoms in the Gaussian Integers: Split, Inert, Ramified as Halve / Stay / Self-Conjugate

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Status. Companion note to *The Genesis Information Project*. An interpretive reading, in resistance variables, of classical facts about the Gaussian integers $\mathbb{Z}[i]$ — the extension flagged as open in Paper 3 §8.5. Contains **no new number theory**: all arithmetic is classical (Fermat, Gauss/Jacobi, the splitting law in $\mathbb{Q}(i)$, Niven) and verified by execution. The contribution is the resistance/genesis reading and the layer-placement.

Role. Captures the Gaussian findings of the June 2026 session. It is a *completed-phase-layer* result — number theory proper — and is **banked as a confirmed landmark, not as upstream genesis structure**, the same standing as the Euler-product reading. Cross-refs: master §4 (the resistance plane; “ i is $\sqrt{\text{NOT}}$ ”; “NOT gives the stage, not the actors”), §6 (the two π ’s; the all-at-once toll), §2.1 (the two NOTs), Axiom 5 (virtual/actual).

0. The setup, in one move

Passing from $\mathbb{Z}$ to $\mathbb{Z}[i]$ is, in the framework’s ontology, a single act: **admitting $\sqrt{\text{NOT}}$ into the number system as an element**, since $\mathbb{Z}[i] = \mathbb{Z}[\sqrt{-1}] = \mathbb{Z}[\sqrt{\text{NOT}}]$. The whole note answers one question — *what happens to each prime-atom once the system can carry a quarter-turn of phase*. (Reading; the operator-as-element identification is master §2.1/§4.)

1. The circle is an iso-resistance contour (Reading)

A Gaussian integer $z = a + bi$ has resistance $\ln |z| = \frac{1}{2} \ln(a^2 + b^2) = \frac{1}{2} \ln n$. So the circle of radius $\sqrt{n}$ — the locus 3Blue1Brown’s lattice-point count lives on — is the level set $\{\text{resistance} = \frac{1}{2} \ln n\}$ in $\mathbb{Z}[i]$. Under $E = \ln |z|$ the growing circle becomes a single horizontal line sweeping upward, and $r_2(n)$ (the sum-of-two-squares count) is the number of Gaussian-integer points that line strikes. The natural “resistor” coordinate is the norm $N(z) = z\bar{z}$, which is multiplicative, so $\ln N$ is additive — the Paper 1 isomorphism extends to $\mathbb{Z}[i]$ through the norm unchanged.

2. The trichotomy (Lead; bottoms out in the two-squares Fact)

Read $r_2(n) = 4 \sum_{d|n} \chi_4(d)$ in resistance language and the three behaviours of a prime become three fates of its atom $\ln p$:

- **Inert** ($p \equiv 3 \pmod{4}$): the atom **stays whole**, $\ln p$, with no conjugate partner — a “real” atom. It survives the count only at even multiplicity. Angle 0 (the unit-1 direction).
- **Split** ($p \equiv 1 \pmod{4}$): the atom **halves**. $\ln p \rightarrow \frac{1}{2} \ln p + \frac{1}{2} \ln p$, the two halves carried by π and $\bar{\pi}$ — *identical magnitude, opposite phase* $\pm \arg \pi$. Master §4 verbatim: “phase is real but

unseen; resistance is phase-blind.” A magnitude probe cannot tell π from $\bar{\pi}$ — one degenerate resistance level.

- **Ramified** ($p = 2$): the atom is a **perfect square**. $2 = -i(1+i)^2$, and the prime above it is, unit-normalised, $e^{i\pi/4} = \sqrt{i} = \text{NOT}^{1/4}$ — the bisector of i ’s phase, the half of the half-turn. Self-conjugate (its “conjugate” is its own associate); invisible to the count ($\chi_4(2) = 0$), so multiplying n by 2 leaves r_2 unchanged.

Verified by execution (June 2026). $r_2(5) = r_2(10) = r_2(20) = 8$ (the 2-factor invisible); $r_2(25) = 12$; $(1+i) = \sqrt{2} e^{i\pi/4}$, $(1+i)^2 = 2i$, $-i(1+i)^2 = 2$; for split p , $\ln |\pi| = \ln |\bar{\pi}| = \frac{1}{2} \ln p$ with opposite phase.

3. The superposition reading (Axiom 5)

A resistance probe at level $\frac{1}{2} \ln 25 = \ln 5$ actualises the modulus: it reads **5**. Which Gaussian factorisation produced that level — $(2+i)(2-i)$, which phase assignment — is conserved by the structure but unreadable by a magnitude probe, hence **virtual relative to that probe** (Axiom 5). $\sqrt{25}$ is what the resistance probe actualises; the conjugate-circuit identity stays virtual until a phase probe pays to read it (Axiom 5: “conversion is never free”).

On “parallel circuitry”: in the *magnitude* coordinate the conjugate factors are in **series** ($\frac{1}{2} \ln p + \frac{1}{2} \ln p = \ln p$); what is parallel-like is the **phase** ($\pm\theta$ recombining to phase 0). A split prime is a **phase-balanced pair** — the pair-version of double-NOT coming home (master §2.1), conjugation being to a general angle θ what double-NOT is to $\theta = \pi$. Distinct from quantum superposition proper, which needs the wave layer (Papers 4–7); χ_4 is one of those characters — the one place the wave layer reaches into this count.

4. The two orthogonalities — commensurate vs incommensurate (Fact at core)

$\sqrt{\text{NOT}}$ gives **4** orthogonal directions in a **2-dimensional** plane — it cycles after four (the units $\{1, i, -1, -i\}$, finite). The primes give **infinitely many** $\mathbb{Q}$ -independent directions in **infinite-dimensional** space (the $\ln p$, independent for free from unique factorisation). “Primes are orthogonal” and “orthogonality is $\sqrt{\text{NOT}}$ ” are both true — *of different orthogonalities*.

The angular anatomy makes the seam exact (Niven’s theorem): the **only** Gaussian prime whose argument is a rational multiple of π is the one over 2, at $\pi/4$; every $p \equiv 1$ prime sits at an angle $\pm \arctan(b/a)$ **incommensurate** with π . So $\mathbb{Z}[i]$ ’s phase axis carries two populations: the **alphabet** (commensurate — the units, χ_4 , the ramified prime — the *stage*, given by $\sqrt{\text{NOT}}$) and the **prime-angles** (incommensurate — the split primes — the *actors*, handed over by which Gaussians are irreducible). Master §4’s stage/actors split, reappearing intact in the phase dimension.

5. The two π ’s, visible together (Lead; connects master §6, §9)

$r_2 = 4 \sum \chi_4(d)$ factors by face: the **4** is the four units = powers of $\sqrt{\text{NOT}}$ (phase quanta); the $\sum \chi_4(d)$ is a character sum, $\chi_4 \in \{\pm 1\}$ = the minimal NOT-sign (the squarefree/phase face); the **average** of r_2 over a growing disc is the area, a real π born only at the all-at-once limit. So the two π ’s are simultaneously present and visibly distinct in $\mathbb{Z}[i]$: the phase π lives in the units and χ_4 ; the magnitude π emerges only on averaging. The seam is exact — $1 - \frac{1}{3} + \frac{1}{5} - \dots = L(1, \chi_4)$, the L -function of precisely the character the wave layer (Paper 4) computes with. The two-squares problem is where the resistance face (r_2 = norm degeneracy) and the phase face (χ_4) of the same Gaussian object meet.

6. Layer placement (added after the proto-phase / proto-pi session)

This note is quantitative throughout — completed-phase layer. By the session’s layer discipline it is **downstream**, banked as a landmark, not upstream genesis structure (same standing as the Euler-product reading). The parts that *are* genesis-native and could promote upward: the **units as the order-4 refinement of the proto-phase alphabet** (master §4, the roots of unity as rational fractions of $2\pi i$), and the **ramified prime at the bisector** $\text{NOT}^{1/4}$ — both live on the commensurate “stage.” The trichotomy proper, the norms, and the χ_4 computation are completed-layer and stay here. The early §4 finding (commensurate alphabet vs incommensurate prime-angles) was the proto-phase/phase distinction before it had been named.

7. Honesty boundary

All arithmetic is classical: Fermat’s two-square theorem, the χ_4 divisor formula (Gauss/Jacobi), the splitting law in $\mathbb{Q}(i) = \mathbb{Q}(\zeta_4)$, Niven’s theorem, the units of $\mathbb{Z}[i]$. None is ours. The contribution is the resistance/genesis reading and the layer-placement. No new leverage on any open arithmetic question is claimed. Verification script archived with the session.

Originating observation (the Gaussian conjugate-circuit framing and the $\sqrt{\text{NOT}}$ reading) is Alexander Pisani’s; the formalisation, the resistance reading, the verification, and this compilation were developed jointly with Claude (Anthropic). June 2026.